

Explaining Flavivirus Congenital Neurotropism with Thermodynamics

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Flaviviruses are a Global Health Threat

- FV include human pathogens like YFV, DENV, JEV, WNV, TBEV
- Typically transmitted by mosquitoes and ticks
- Millions of infections every year
- Broad range of clinical manifestations
 - Headache / rash / (hemorrhagic) fever / meningitis / encephalitis
- FV neurotropism reported in adults for DENV, YFV, WNV

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- 2015-2017 ZIKV outbreak in the Americas
- Congenital neurotropism
- High increase in microcephaly cases in newborns



Source: Wikipedia

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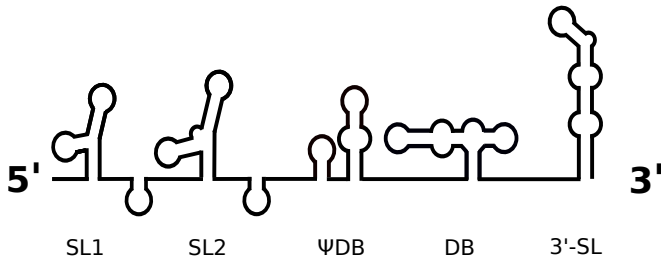


Source: Wikipedia

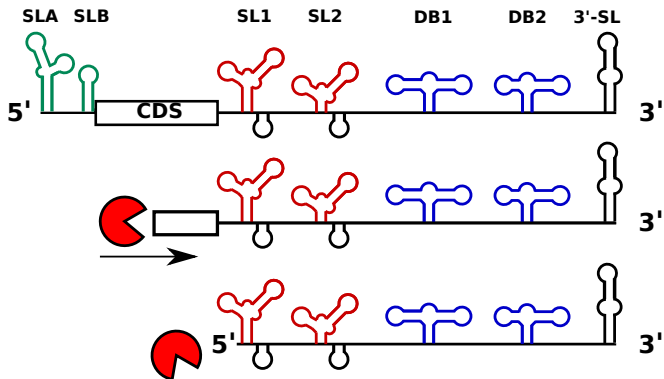
Are there other emerging FV that have a similar neurotropic potential in the developing fetus?

Flavivirus genome characteristics

- Single-stranded, positive-strand RNA viruses
- Enters cell through receptor-mediated endocytosis
- Capped, non-polyadenylated genome (gRNA) of 10-12kb length
- Encodes a single ORF, flanked by structured 5'-UTR and 3'-UTR
- Translation of FV ORF yields a single polyprotein



Flaviviruses hijack the host mRNA degradation machinery



- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial Xrn1 degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures
- sfRNA modulates pathogenicity
- Many host proteins bind sfRNA

Musashi interacts with 3'UTR

Musashi (Msi) is a highly conserved family of proteins in vertebrates and invertebrates that act as a translational regulator of target mRNA.

- Involved in cell proliferation and differentiation
- Two paralogs in mammals, Msi-1 and Msi-2
- Expressed in stem cells and overexpressed in tumors
- Two RNA-regulation domains (RRM)
- Binds single-stranded UAG core motif in the 3'UTR of mRNA

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A Conserved Three-nucleotide Core Motif Defines Musashi RNA Binding Specificity*

Received for publication, July 16, 2014, and in revised form, October 20, 2014. Published, JBC Papers in Press, November 3, 2014, DOI 10.1074/jbc.M114.597112

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Mol Cell Biol 35:2965-2978 (2015)

RNA-Binding Protein Musashi1 Is a Central Regulator of Adhesion Pathways in Glioblastoma

Philip J. Uren,^a Dat T. Vo,^{b,c} Patricia Rosa de Araujo,^{b,c} Rebecca Pötschke,^d Suzanne C. Burns,^b Emad Bahrami-Samani,^a Mei Qiao,^b Raquel de Sousa Abreu,^b Helder I. Nakaya,^{e,f} Bruna R. Correa,^b Caspar Kühnöl,^d Jernej Ule,^g Jennifer L. Martindale,^h Kotb Abdelmohsen,^h Myriam Gorospe,^h Andrew D. Smith,^a Luiz O. F. Penalva^{b,c}

Musashi is involved in ZIKV neurotropism

A recent study has shown that Msi-1 promotes ZIKV replication in glioblastoma, neuroblastoma and neuronal stem cells

Science

REPORTS

Cite as: P. L. Chavali *et al.*, *Science*
10.1126/science.aam9243 (2017).

Neurodevelopmental protein Musashi 1 interacts with the Zika genome and promotes viral replication

Pavithra L. Chavali,^{1*†} Lovorka Stojic,^{1*} Luke W. Meredith,² Nimesh Joseph,¹ Michael S. Nahorski,³ Thomas J. Sanford,² Trevor R. Sweeney,² Ben A. Krishna,⁴ Myra Hosmillo,² Andrew E. Firth,² Richard Bayliss,⁵ Carlo L. Marcelis,⁶ Susan Lindsay,⁷ Ian Goodfellow,² C. Geoffrey Woods,³ Fanni Gergely^{1‡}

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Another study addressed the question whether related arboviruses could cause transplacental infection and fetal demise

SCIENCE TRANSLATIONAL MEDICINE | RESEARCH ARTICLE

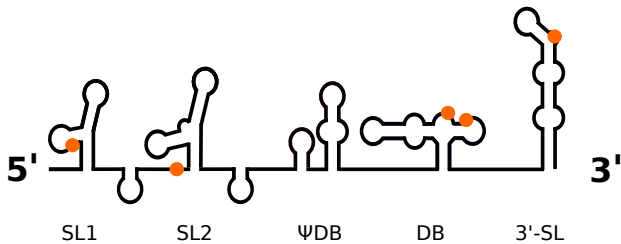
EMERGING INFECTIONS

Zika virus–related neurotropic flaviviruses infect human placental explants and cause fetal demise in mice

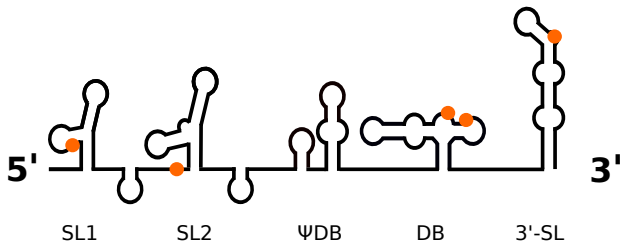
Derek J. Platt,^{1*} Amber M. Smith,^{2*} Nitin Arora,^{3,4} Michael S. Diamond,^{1,2,5,6}
Carolyn B. Coyne,^{3,4} Jonathan J. Miner^{1,2,5†}

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ZIKV 3'UTR revisited



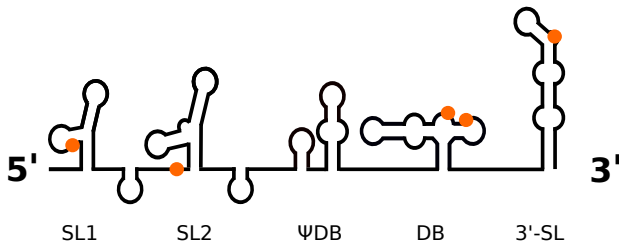
ZIKV 3'UTR revisited



$$Z = \sum_s e^{-E(s)/RT}$$

$$\Delta G_{\text{open}} = -RT \ln P(\text{unpaired})$$

ZIKV 3'UTR revisited

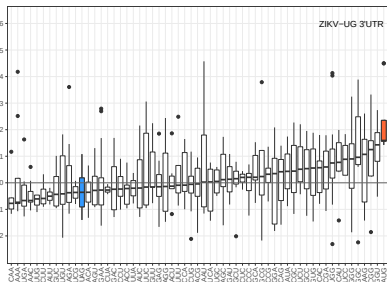
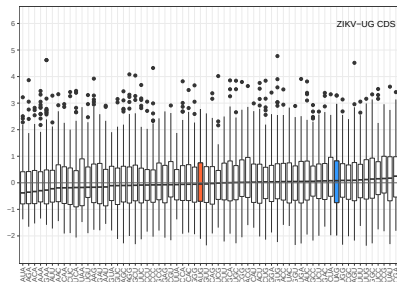
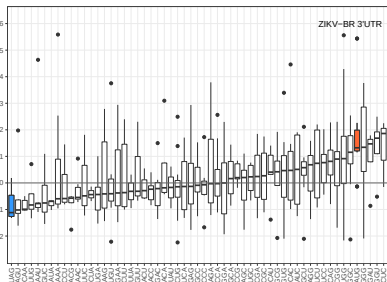
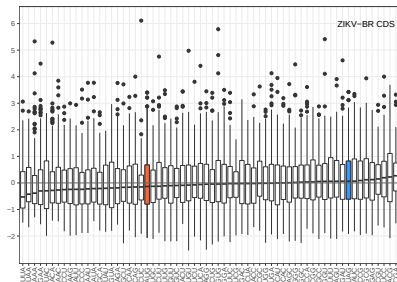


$$Z = \sum_s e^{-E(s)/RT}$$

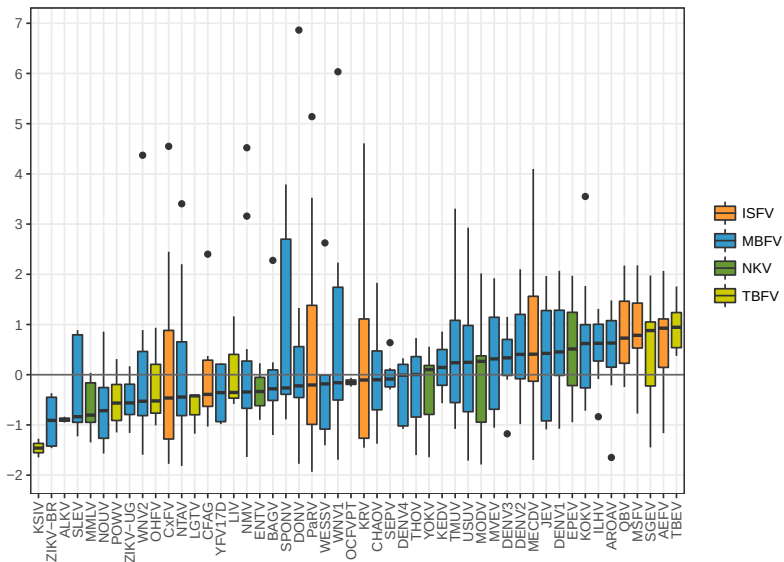
$$\Delta G_{\text{open}} = -RT \ln P(\text{unpaired})$$

We use `RNAplfold` to compute the opening energy of trinucleotide x in a genomic (± 100 nt) as well as a shuffled sequence context and compute a z score statistics.

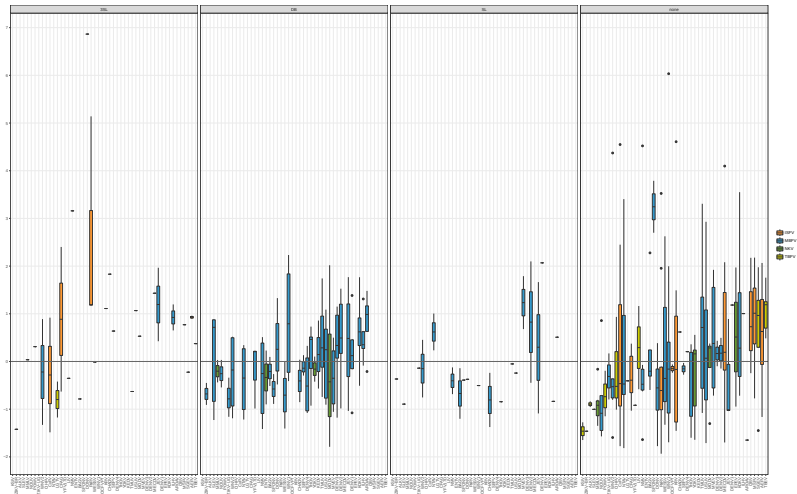
Trinucleotide opening energies in ZIKV



MBE opening energies in related viruses

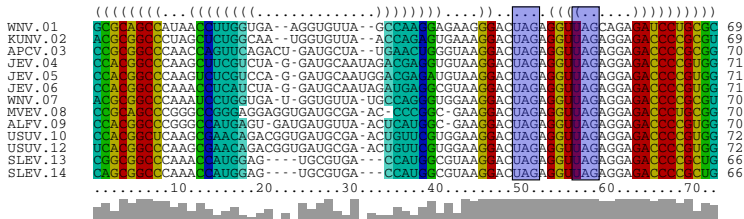
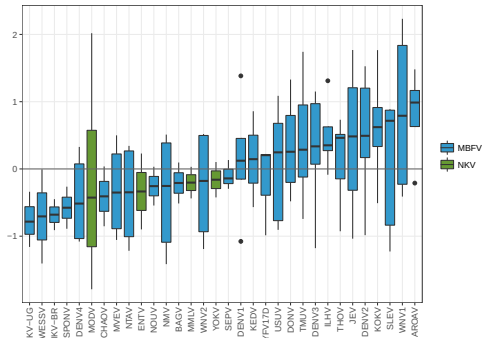


MBEs opening energies by xrRNA



MBEs accumulate within dumbbell (DB) elements in MBFV and NKV

MBE in MBFV and NKV DB elements



Structure and sequence conservation in DB elements

			((((((((.....(((.....)))))).....)))).....))))))	
YORKV_NC_005039.1	DB.1	211-292	GGGAGGCCUCCGCC---AAUGGUGGCUUUAUAUUAUAGGACUACGCAUUGGUGCAUUGGGGACUAGCGGUUAGAGGAGAGACCCUCC	82
DONV_NC_015997.1	DB.1	140-221	AGCGGGCCAGAUU---UCUGUUCUCA---AGCUGAGGACACUGAGCACUC---UGGAAGUAGAGGAGAAACCCGCCA	77
CHAOV_NC_017086.1	DB.1	130-201	AGGGGGACACAUCC--C-ACCAGC---AGCCCGAGCUGGACAAGGCA-UG---UGUAUACGCGGUUAGAGGAGAGACCCGCCCA	72
SEPV_NC_008719.1	DB.1	262-333	AGGGAAGCCACUCCG--CUGUCUG---GUCGCCGCGCAGC---AGGAAAGAAGGGGCUUAGAGGUUAGAGGAGAGACCCUCCG	72
WESSV_NC_012735.1	DB.1	274-345	UGGAGAGCCUCCAC--CCACCGU---UAACGCGCGAGUGU---GGGAAGAUGGGGCUUAGAGGUUAGAGGAGAGACCCUCC	72
YFV17D_NC_002031.1	DB.1	1307-379	ACGAGGAGCCUCCG--CAGACACC---UCCACCGUGUGUG---UAGAAGAGCGGGGCUUAGAGGUUAGAGGAGAGACCCUCCAG	73
USUV_NC_006551.1	DB.1	386-457	CCAGCGGCCAAGC---GAGACAGC---GUUGUGGACUGUG---UCG---UGGAAGC-ACUAGAGGUUAGAGGAGACCCUCCG	72
MVEV_NC_000943.1	DB.1	340-409	CCCGACCCCGGGG--CGGAGGAG--GUGAUGUCAAAC-CC-CGG--C-GAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCGG	70
JEV_NC_001437.1	DB.1	310-380	CCACGGGCCCAAGC--CUCUGUAG--GAUGCAAU-AGACG--AGG--UGUAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	71
AROA_V_NC_009026.2	DB.1	141-211	GGGCGGCCCAAGC--CUGGUAUG--AACCGUUG-AUCCG--AGC--CGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	71
TMUV_NC_015843.2	DB.1	348-416	CCACGGCCUGAA--UGAUGAGC--U--GGUGUGACCAU--CAU---GGAGGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	69
ZIKV-BR_NC_035889.1	DB.1	1238-307	CUGGGGCGUGAAG--UGGAGUCA--G--CUGUGAUCUC--CAG--AGAGGG-ACUAGUGUAGAGGAGAGACCCUCCG	70
ZIKV-UG_NC_012532.1	DB.1	238-307	CCACGGCCUGAAG--UGGAGUCA--G--CUGUGAUCUC--CAG--CAGAGG-ACUAGUGUAGAGGAGAGACCCUCCG	69
SPONV_SA-Ar	DB.1	206-274	GUGGGGCGUGAAG--GCAGAGGA--G--CUGUGACUCU--GCU---GGAGGG-ACUAGCGGUUAGAGGAGAGACCCUCC	69
NMV_NC_032088.1	DB.2	364-428	UGUGGGCCC--AAG--CGAG-U-G--AAGCUGU-AAA-CU-CGU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCC	65
NTAV_NC_018705.3	DB.2	441-505	CAACAGGCC--AAC--CUGG-C-A--AAGCUGU-AAA-CU-CGU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	65
WNV1_NC_009942.2	DB.2	438-503	GCACGGCC--AGC--CUGG-CU-G--AAGCUGU-AGG-UC-AGG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	66
WNV2_NC_001563.2	DB.2	382-447	CCACGGCC--AAC--UUGG-CU-G--AAGCUGU-AGG-CC-AGG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	66
MVEV_NC_000943.1	DB.2	417-483	GAGUGGCC--AAC--CUGG-C-G--AAGCUGUAGAG--CG--GGU---GGACGG-ACUAGAGGUUAGAGGAGAGACCCUCC	67
SLEV_NC_007580.2	DB.2	359-426	CAAGGGCCCAAG--CGC-CUCA--AAGCUGUAGAG--AC--GGG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
DENV4_NC_002640.1	DB.2	201-269	AGGGGGCCCGAAG--CAG-AGG--AAGCUGUACUC--CU--GGU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCC	69
DENV2_NC_001474.2	DB.2	268-335	UGGGGGCCC--AAG--CGCA-GAUG--AAGCUGUAGUC--UC--GCU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
THOV_NC_034151.1	DB.2	359-426	CCACGGCC--AAU--UCUG-UGUG--AAGCUGUAGC--CA--GAG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
DENV1_NC_001477.1	DB.2	278-345	CGGGGGCC--AAC--ACCA-GGGG--AAGCUGUACCC--UG--GUG---GUAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
JEV_NC_001437.1	DB.2	391-458	UAGCGGCC--AAC--CCCC-CUCG--AAGCUGUAGAG--GA--GGU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCC	68
TMUV_NC_015843.2	DB.2	424-491	GCAAGGCC--AAC--CUAG-AGUC--AAGCUGUAUCU--CU--AGG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
USUV_NC_006551.1	DB.2	465-533	UGGGGGCCC--AAG--CCGUUCCG--AAGCUGUAGGA--AC--GGU---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCC	69
ILHV_NC_009029.2	DB.2	195-262	CCACGGCC--AAU--UCUG-UGUG--AAGCUGUAGC--CA--GAG---GGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	68
DENV3_NC_001475.2	DB.2	257-325	CGGGGGCCCGAAG--ACUG-AGGG--AAGCUGUACCU--CC--UUG--CAAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	69
WNV1_NC_009942.1	DB.1	361-429	ACCGGGCCCUUAG--CCC-GGUU--AUGGUGUUAACC-A--GGG--CGAAAG-ACUAGAGGUUAGAGGAGAGACCCUCCG	69
WNV2_NC_001563.2	DB.1	305-373	ACGCGGCCCAUAGC--CU--GCUA--AUGGUGUUAACC-A--GGG--AGAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	69
NMV_NC_032088.1	DB.1	288-358	CCACGGCGUGAGC--ACU-UCUC--UUGCGGAGAGAUU--GUG--UCACCGGGACUAGAGGUUAGAGGAGAGACCCUCCG	71
SLEV_NC_007580.2	DB.2	287-355	CCACGGCC--AA--ACCAAGC--AAGCUGUGACAUU--GC--GAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	66
THOV_NC_034151.1	DB.1	283-348	CCACGGCC--AA--UCUUGG--AGGGCGUGGCCCAA--GA--GUAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	66
NTAV_NC_018705.3	DB.1	366-432	CCCGGGCC--AA--GCAAGG-A--AUGGUGUGCCCCUU--GU---GGAAGG-UCUAGAGGUUAGAGGAGAGACCCUCCG	67
BAGV_NC_012534.1	DB.1	372-438	CUGUGGCC--AA--CAAGGG-A--AUGGUGUGACCCUU--GU---GUAAGG-ACUAGAGGUUAGAGGAGAGACCCUCCG	67
KOKV_NC_009029.2	DB.1	290-355	UGGAGCGCC--GA--GGGGGAGAA--GAACCGGAUCUCC--CC---AAAGG-ACUAGAGGUUAGAGGAGAGAC--CGUGA	66
ILHV_NC_009029.2	DB.1	121-187	CCACGGCC--AAU--UCUGAGGA--CAGCGGACUCC--GAG--GAAGG-ACUAGAGGUUAGAGGAGAGAC--CGUG	67
DENV4_NC_002640.1	DB.1	111-178	CGGAGGCCAUAGC--GCACAGG--AAGCUGU-ACGCGU--GGC--AAUUAUGGACUAGCGGUUAGAGGAGAGACCCUCC	68
DENV1_NC_001477.1	DB.1	194-261	GGGAGGCGUGCAAA--CAUUG--AAGCUGU-ACGCAU--GGG--GUAGCAGACUAGUGGUUAGAGGAGAGACCCUCC	68
DENV3_NC_001475.2	DB.1	173-240	GGGAGGCGUGCAAA--CUGUGG--AAGCUGU-ACGCAU--GGG--GUAGCAGACUAGCGGUUAGAGGAGAGACCCUCC	68
DENV2_NC_001474.2	DB.1	181-248	GGGAGGCGCAAAA--CAUUG--AAGCUGU-ACGCAU--GGC--GUAGUGGACUAGCGGUUAGAGGAGAGACCCUCC	68
KEDV_NC_012534.1	DB.2	197-261	GGGAGGCGCAAGC--CGCAUC--ACUGC--UAA-C--UGGAAGACUAGAGGUUAGAGGAGAGACCCUCCG	65
KOKV_NC_009029.2	DB.2	368-432	GGGAGCC--AU--CUCAGG-A--AAGCUGUACCCUG--GGG--GAAGG-ACUAGAGGUUAGAGGAGAGAC--CUCC	65
AROA_V_NC_009026.2	DB.2	223-288	CCAAAGGCCCAACC--UGUCG--AACCUUAUG--C-AGG--UAAAAG-ACUAGAGGUUAGAGGAGAGACCCUCCG	66
NOUV_NC_033715.1	DB.1	282-352	AGCCGCGCAAAAGGCCCAUCAGCUG--AAGCUUAUAGGUGG--GGG--AAGG-ACUAGAGGUUAGAGGAGAGACCCUCC	66
ENTV_NC_008718.1	DB.1	42-99	CGGAGGCC--C--CGUGC--GAAACAGU--CGACGGGGACUAGAGGUUAGAGGAGAGACCCUCCG	58
MLLV_NC_004119.1	DB.1	200-251	GGGAGGC-C--CGAAG--AAAGCUGA--UAGCGGUUAGAGGAGAGACCCUCC	52
			10.....20.....30.....40.....50.....60.....70.....80.....	

- Highly conserved MBE pair in DB elements
- Central multiloop and distal hairpin loop

- Our biophysical model corroborates experimental findings of Msi involvement in ZIKV replication
- MBEs in flavivirus DB elements show primary sequence conservation
- ZIKV might not be alone in its capacity to cause severe neuropathology in the developing fetus
- A possible role of Msi in the replication cycle of some FV?

Acknowledgments

Collaborator

Adriano De Bernardi Schneider
(UNC Charlotte)



Der Wissenschaftsfonds.

SFB RNA regulation of the
transcriptome (FWF-F43)

TBI Vienna

Ivo L. Hofacker
Roman Ochsenreiter
Andrea Tanzer



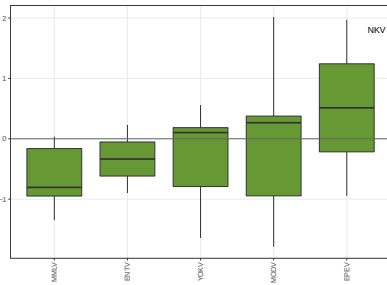
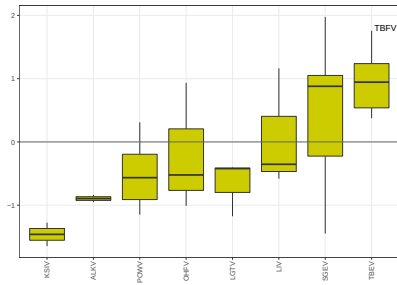
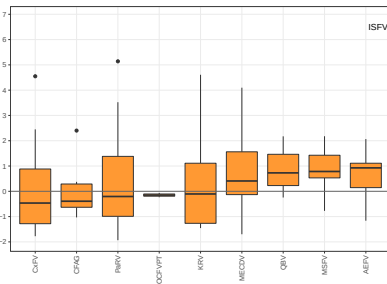
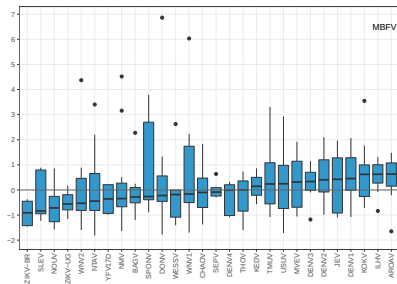
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Thank you!

MBE opening energies in related viruses



MBE opening energies in related viruses

